Symmetric residue budgets for a 258-variable consistency quartic

A direct equation for Con(ZFC), with compact companions

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Abstract

We give a fixed polynomial $P_{258} \in \mathbb{Z}[x_0, \dots, x_{257}]$ of exact degree four such that ZFC proves that P_{258} has no integer zero if and only if ZFC is consistent. The equation is supplied completely as a sum of 186 explicitly printed quadratic residual squares and as all 5,319 collected coefficients. Its coefficient height is 47,281. It refines the latest inspected direct numerical-proof-calculus construction, which has 282 variables, 192 residuals, 5,491 terms and height 128,446.

The main change replaces eleven private interval certificates by shared quadratic bounds on signed remainders. A radix estimate proved before the factorial fixed point makes congruence uniqueness recover the missing signs. Two adjacent scales can share a bound. A compatible odd-square radix replaces a further sign certificate, while a changed recovery order avoids circular reasoning. Thirty elementary residual-row operations then reduce coefficient height without changing the zero set.

A companion quartic has 261 variables and 185 residuals. Two separate ten-positive-variable equations save one arithmetic operation each, giving 77 operations at degree 630 and 78 at degree 468. Their proof removes an auxiliary positivity shift by excluding a low Pell index. The new ordinary proofs compose with explicitly retained repository calibrations. No global simplicity minimum or new end-to-end proof-assistant certificate is claimed.

1 The exact result and the comparison boundary

The comparison snapshot is ProveIt [1] commit `e0c98c629bf10d30b695884f6b1d2ab361bdf513`. We inspected the two requested research areas and all six ZIP archives then present in `docs/incoming`. In particular, the comparison below includes the incoming round-17 cofinal-anchor construction [2]; it is not a comparison against an obsolete earlier collection overview.

Theorem 1.1 (The supplied direct quartic)**.** *Let $r_0, \dots, r_{185} \in \mathbb{Z}[x_0, \dots, x_{257}]$ be the literal quadratic polynomials in Appendix A, and define*

$$\boxed{P_{258}(x_0, \dots, x_{257}) = \sum_{j=0}^{185} r_j(x_0, \dots, x_{257})^2 = 0.} \tag{1}$$

All 258 coordinates range independently over $\mathbb{Z}$. Composing the proofs in this report with the retained numerical C222/refutation calibration,

$$\mathrm{ZFC} \vdash \Big(\neg\exists \mathbf{x} \in \mathbb{Z}^{258}\ P_{258}(\mathbf{x}) = 0\Big) \longleftrightarrow \mathrm{Con(ZFC)}. \tag{2}$$

The polynomial has exact degree four, 5,319 nonzero collected monomials, and coefficient height 47,281. Every coordinate is live. Its coefficient of x_0^4 is 13, and that of $x_{10}^2 x_{12}^2$ is 47,281.

Here consistency means the absence of a finite syntactic refutation in the repository's specified ZFC proof system. Equation (2) is the result; it is not an independent assertion that either side is true. The actual mathematical dependency is an ordinary proof formalizable in ZFC. This packet does not supply a newly checked internal ZFC proof object or a new complete Lean/Rocq build of the whole chain.

Direct equation over $\mathbb{Z}$	Variables	Rows	Degree	Terms	Height
Incoming matched parent	282	192	4	5,491	128,446
Shared four-budget companion	261	185	4	5,332	47,281
Main odd-square equation	**258**	**186**	**4**	**5,319**	**47,281**

The 258-variable result removes 24 variables and six rows from the matched parent, removes 172 collected terms, and reduces its height by about 63.2%. The 261-variable companion uses one fewer residual than the main equation. Both have literal coefficient data and reconstruction checks. These are achieved improvements in a specified direct lineage. Existing native-encoding equations have fewer variables, sometimes at much higher degree or with very large fixed coefficients. We do not call this a global variable record or a globally simplest polynomial.

The direct branch encodes the numerical C222 proof calculus. It does not simulate K215, specialize Jones's universal polynomial, or hide a machine description in a large coefficient port. Its final coefficients are the ordinary integers printed by its residuals. The separate compact branch in Section 8 retains its native/K215 ancestry and must be assessed on its own domain and size metrics.

2 The inherited arithmetic interface

The parent direct construction is explicit and substantial. We retain its complete guarded quadratic source, its source metadata, the ordinary proofs of its arithmetic front, and the named logical interfaces. The new proof changes a narrowly identified part of that source rather than claiming to reprove the whole compilation from a finite test.

The logical chain is

$$\begin{aligned} \text{integer zero} &\iff \text{finite C222 derivation} \\ &\iff \text{finite ZF refutation} \\ &\iff \text{finite ZFC refutation.} \end{aligned}$$

The last two retained interfaces are:

- `KEndpoint.c222_iff_zf_refutation`;
- `ConstructibleChoice.zf_refutation_iff_zfc_refutation`.

Their files are included as source evidence; inclusion is not a claim that their entire transitive proof-assistant dependency tree was rebuilt.

The arithmetic construction packs finite local proof data by Chinese remaindering, interprets binomial divisibilities as range conditions, and uses explicitly bounded Pell certificates to represent powers. Its signed orientation stage recovers both coordinates of each principal Pell pair before it uses the low-digit remainder signs. This order is the opportunity exploited here.

Write t, F, B, x, y, V_0, q for its semantic arithmetic values and $a_1, \dots, a_9$ for its actual CRT representatives. They are polynomials in the source graph's coordinates, not new variables in (1). At the stage after power recovery but before the factorial fixed point, the parent proves

$$t \geq 16, \quad F \geq 1, \quad B \geq F + 1 \geq 2, \qquad V_0 = (F^2+1)^t, \qquad y = V_0(V_0 - 1), \tag{3}$$

$$x \geq 2 + (t+1)y, \qquad a_i \geq 0, \qquad H_0 = 1 + \sum_{i=1}^{9} a_i \geq 1, \qquad q = (BH_0)^x. \tag{4}$$

In particular, this stage does not yet use $F = t!$. The raw ceiling is $FB = V_0 + s$ with $s \geq 0$, and its two endpoint congruences give the large lower bound on x. A modified order for the odd-square version will be proved in Section 6.

There are eleven modular numerator requests. Each has a denominator $D_i = q^{k_i}$ and a low remainder r_i. The old rows impose

$$D_i = r_i + s_i + 1, \qquad r_i s_i = u_i^2 + v_i^2 + w_i(w_i + 1). \tag{5}$$

Once $D_i > 0$ is known, the second row makes r_i, s_i have the same sign, and the first forces them both nonnegative. It costs three private integer coordinates per request. We remove precisely the eleven product rows in (5); all the first rows are retained. The three initial ceiling/output sign graphs are separate from these eleven rows.

3 A residue-recognition lemma with a shared bound

The useful fact is not an unrestricted replacement for arbitrary interval guards. It relies on the intended residues occupying a small part of their interval.

Lemma 3.1 (Symmetric residue recognition)**.** *Let $D \geq 1$ and let $\rho_1, \ldots, \rho_m$ be integers with $0 \leq \rho_i < D/8$. Suppose integers r_i satisfy $r_i \equiv \rho_i \pmod{D}$ and*

$$D^2 - 4\sum_{i=1}^{m} r_i^2 = 1 + u^2 + v^2 + w(w+1). \tag{6}$$

Then $r_i = \rho_i$ for all i. Conversely, if $m \leq 7$, the canonical choice $r_i = \rho_i$ admits integers u, v, w satisfying (6).

Proof. For every integer w, $w(w+1) \geq 0$. Equation (6) implies $|r_i| < D/2$. Thus

$$|r_i - \rho_i| < D/2 + D/8 = 5D/8 < D.$$

The difference is a multiple of D, so it is zero.

For the converse, $4\sum_i \rho_i^2 < 7D^2/16 < D^2$. Therefore the integer $D^2 - 4\sum_i \rho_i^2 - 1$ is nonnegative. Every nonnegative integer n has a representation $n = u^2 + v^2 + w(w+1)$. Indeed, the three-square theorem applied to $4n+1$ gives exactly one odd coordinate and two even coordinates; division by four gives the representation. This standard ternary form is also used in the parent and in current formalized Diophantine constructions [5]. □

The lemma allows the r_i to be signed at the moment their congruences are checked. Nonnegativity is a conclusion. Its applicability here rests on proving the bound $\rho_i < D/8$ at the right point in the argument.

4 The radix margin is available before factorial recovery

The parent constructor ledger supplies the positive integer $\ell = 2^{22} = 4,194,304$ and the derived-label bound

$$h \leq (n_* + 1)^{\ell}, \qquad n_* = \max(1, a, b, c, d),$$

where a, b, c, d are four of the nine nonnegative CRT representatives. Since $n_* + 1 \leq 2H_0$ and $B \geq 2$, we can use

$$h \leq (2H_0)^{\ell} \leq (BH_0)^{\ell}. \tag{7}$$

This deliberately permits $H_0 = 1$. We do not use the later proof that a particular encoded index is nonzero.

By (3), $V_0 \geq 2^t$, so

$$y \geq 2^t(2^t - 1) > \ell, \qquad y \geq t + 2 \qquad (t \geq 16).$$

Together with (4), these imply the two coarse but sufficient integer bounds

$$x \geq \ell t + 4, \qquad x \geq t(t + 1) + 4. \tag{8}$$

Lemma 4.1 (Uniform low-digit bound)**.** *For every one of the eleven numerator requests and every binomial coefficient below its requested digit position,*

$$\binom{n}{j} \leq q/16. \tag{9}$$

This conclusion follows from the pre-factorial facts (3)–(4) *and the constructor ledger.*

Proof. There are three cases. For an ordinary representative $0 \leq n < H_0$, the denominator exponent k is at most x. If $0 \leq j < k$, then

$$\binom{n}{j} \leq \max(1, n)^j \leq H_0^x \leq q/16,$$

since $q = B^x H_0^x$, $B \geq 2$ and $x \geq 4$. This includes $n = 0, j = 0$.

For the derived-label representative, $k = t + 1$ and $j \leq t$. Equations (7) and (8) give

$$\binom{h}{j} \leq h^j \leq (BH_0)^{\ell t} \leq (BH_0)^{x-4} \leq q/16.$$

For the factorial upper index $n = F^2 + t$, we have $F^2 + t \leq B^{t+1}$: indeed $F^2 < B^2 \leq B^t$, $t \leq 2^t \leq B^t$, and $2B^t \leq B^{t+1}$. Therefore for $j < t$,

$$\binom{F^2 + t}{j} \leq (F^2 + t)^t \leq B^{t(t+1)} \leq B^{x-4} \leq q/16.$$

None of the three arguments assumes $F = t!$. □

For $D = q^k$, define the canonical low residue in the proof by

$$\rho(n, k) = \sum_{j=0}^{k-1} \binom{n}{j} q^j.$$

This is proof notation only; it is not a variable-length operation in the final polynomial. The binomial theorem gives $\rho(n, k) \equiv (q + 1)^n \pmod{D}$, and Lemma 4.1 gives

$$0 \leq \rho(n, k) \leq \frac{q(D - 1)}{16(q - 1)} \leq \frac{D - 1}{8} < \frac{D}{8}. \tag{10}$$

The retained numerator equations make each actual r_i congruent to this canonical residue modulo its positive denominator. The factorial numerator has a high part $D(B + qv)$; a range numerator has high part DNv. Both are multiples of D regardless of the still-unproved sign of v. This is why the congruence can be used before the digit signs.

5 Pooling, merging, and soundness of the direct graph

The eleven requests have four denominator levels:

Denominator	Requests	Semantic upper indices
q^t	2	factorial upper index and encoded index
q^{t+1}	5	four local inputs and derived label
q^{y+1}	2	two local state representatives
q^x	2	two strict beta quotient representatives

For each level, introduce one row (6) instead of its old private product guards. The four rows use twelve guard coordinates in place of thirty-three. After the inherited exact-power recovery, Lemma 3.1 and (10) prove every $r_i = \rho_i \geq 0$. The retained linear rows then give $s_i = D_i - 1 - r_i \geq 0$. Thus every deleted old product guard can be restored using a ternary representation of $r_i s_i$.

All later inherited steps now apply. They authenticate the factorial digit, obtain $F = t!$ from the ceiling fixed point, decode the amplified CRT range conditions, and extract a finite C222 derivation from the unit-weight proof-flow certificate. This proves soundness. For completeness, start with the parent's witnesses for any finite refutation; their canonical low residues satisfy (10), and the converse of Lemma 3.1 supplies the four new budgets. The exact compiled result has 261 variables and 185 rows.

5.1 Merging adjacent powers of the radix

Let $D = q^t$ and $D' = qD = q^{t+1}$. Introduce just two product coordinates

$$z_0 = qr_0, \qquad z_1 = qr_1$$

for the two lower-level remainders. Pool z_0, z_1 together with the five higher-level remainders under denominator D'. The seven-entry budget forces $|z_i| < D'/2$, hence $|r_i| < D/2$ because $q > 0$. The original denominator congruences therefore recover the same lower-level residues. Conversely, $q\rho_i < D'/8$ and the other five canonical residues have the same bound, so the seven-entry budget has witnesses.

This replaces six guard coordinates by three and adds two product coordinates, saving one variable. Its 260-variable intermediate has 186 rows. The polynomial expressions are still quadratic before taking their sum of squares.

6 An odd-square radix and a different recovery order

The parent has one initial scalar sign guard on the factorial digit's output slack o, with $q = B + o + 1$. Remove that three-coordinate guard and add the single quadratic row

$$q = (2z + 1)^2, \qquad z \in \mathbb{Z}. \tag{11}$$

This saves two more variables. It is a restriction on admissible raw witnesses, so it requires a new completeness argument as well as soundness.

6.1 Soundness without an initial lower bound of two on the radix

Initially (11) gives only $q \geq 1$. The signed semantic anchor and square-frame argument orient all principal Pell pairs without using the deleted sign of o. Interpret $V_0 = (F^2 + 1)^t$ first: its exponent

$t \geq 16$ is already known, and the base-one case is allowed. The two retained ceiling signs then give $F \geq 1$, $B > F$, and $y = V_0(V_0 - 1)$.

Now use the existing request $N = B^x$. Since $B \geq 2$, the signed-power lemma forces $x \geq 0$. The first beta endpoint gives $x \geq F + 1 \geq 2$, and the last gives $x \geq 2 + (t + 1)y$. The numerator requests have base $q + 1 \geq 2$, so their exponents are nonnegative actual representatives; the anchor index is nonzero. Consequently $H_0 \geq 1$. The request with base $BH_0 \geq 2$ then gives

$$q = (BH_0)^x \geq B^2 > B.$$

Thus $o = q - B - 1 \geq 0$ is recovered. From this point, all previous power and digit arguments apply with $q \geq 2$. The order is

$$q \geq 1 \;\Rightarrow\; V_0 \;\Rightarrow\; B \geq 2 \;\Rightarrow\; x \geq 2 \;\Rightarrow\; H_0 \geq 1 \;\Rightarrow\; q \geq B^2 > B.$$

No step uses the very sign it is meant to establish.

6.2 Completeness by compatible parity choices

Begin with any finite refutation and the parent's cofinal construction. Choose its horizon $t \geq 16$, $F = t!$, $V_0 = (F^2 + 1)^t$ and $y = V_0(V_0 - 1)$. The number y is even, so every beta CRT modulus $1 + (i + 1)y$ is odd. The beta code x can therefore be chosen even and arbitrarily large in its required residue class. Its endpoint and quotient conditions are preserved by that choice.

The factors $d_i = 1 + F^2/i$ are odd: $i \mid F$ and $F^2/i = F(F/i)$ is even. Therefore $B = \prod_i d_i$ and $N = B^x$ are odd. The independent representative CRT moduli have odd product N^2, so each representative's parity can be prescribed independently.

Prescribe opposite parities to a, b, as required to make the semantic anchor h even. Use one of the other seven representatives to make the sum of all nine representatives even, hence H_0 odd. Ordinary representatives remain in $(0, 2N^2]$. The dominant representative is chosen as $a_0 + 2N^2$ or $a_0 + 3N^2$, maintaining the required parity and $2N^2 \leq a < 4N^2$. All inherited dominance and cap estimates remain valid.

Now BH_0 is odd and x even, so

$$z = \frac{(BH_0)^{x/2} - 1}{2} \in \mathbb{Z}$$

supplies (11). The shared remainder budgets can be completed as before. This proves completeness of the 258-variable graph.

7 Smaller coefficients by reversible residual operations

The preceding graph is converted into literal quadratic residuals by the retained Horn factorization and integral monic coordinate elimination. Every eliminated coordinate has an explicit polynomial restoration. Integral coordinate translations are bijective over the integer domain. After those steps the main equation has 258 variables, 186 rows, 5,319 terms and height 128,446.

We then apply 30 elementary transformations to its ordered residual list:

$$r_i \longleftarrow r_i + kr_j \qquad (i \neq j, k \in \mathbb{Z}). \tag{12}$$

Each is a unimodular row transformation, with inverse obtained by replacing k by $-k$. Therefore it preserves the simultaneous zero set of the residuals over every commutative ring. Over $\mathbb{Z}$ the sum of their

squares vanishes exactly when each residual vanishes, so the transformed single equation has exactly the same integer zero set.

The search for useful operations was finite and heuristic. Its output, not the search heuristic, is the certificate: every operation and its reverse is supplied and checked exactly. These operations lower the largest residual coefficient from 256 to 128 and the collected quartic height to 47,281. The final support count returns to 5,319. No claim of optimal row basis, optimal integer translation, or globally minimal coefficient height is made.

This is a useful separation of concerns. Number theory justifies deleting sign witnesses. Reversible algebra improves the displayed coefficients. Neither improvement is inferred from testing whether the final equation has a solution.

8 A compact companion: deleting the auxiliary shift

Two compact companions have ten strictly positive integer unknowns. They retain the entire fixed indexed BFS943 coefficient tuple of the incoming `ConZFC_Certificate_Reductions_2026-10-25.zip` at the repository snapshot used in this report. Their live binary arithmetic costs and exact degrees are

$$(77, 630) \qquad \text{and} \qquad (78, 468).$$

Each saves one addition at unchanged degree from its matched $(78, 630)$ or $(79, 468)$ parent. This is a small arithmetic reduction. The new reasoning is a low-index exclusion that permits signed transport before the input coordinates have acquired their intended interpretation. It proves that a previous shift $T + \alpha$ was redundant.

8.1 The complete polynomial

The only independent unknowns are

$$J, F, \alpha, i, T, w, s, n, \eta, \zeta \in \mathbb{Z}_{>0}.$$

The four fixed integers β, K_0, M_C, M_F are specified by the finite recipe below. Every other symbol in the following display is a polynomial abbreviation, and adds no unknown:

$$\begin{gathered}
q = \beta J + 1, \quad h = q - 1, \quad C = q - F - \alpha, \quad X = wq, \\
Y = q^3 s, \quad k = \eta + \zeta, \quad c = kY + \eta, \quad D = Y^2, \\
A = Y(X + 1) + 1, \quad \Delta = A^2 - 1, \quad H = 4A - 5, \\
V_0 = XD, \qquad E = 1 + V_0(V_0 + 1)k^2, \\
R = [h(q - F) + \alpha](q^2 - 1) + (M_C + qM_F)J, \\
e = k - R - 1, \quad B = (K_0 + w)C - F, \quad U = ic^2, \\
a = \Delta U^2 E, \quad F_0 = 1 + a, \quad G = EF_0, \\
F_* = 1 + 4aF_0, \quad Q = \Delta(F_* - 1), \\
b = cT - R, \quad V = F_* b - c, \quad K_+ = X(3c - 2X) + 2.
\end{gathered}$$

For the 77-operation equation choose $L = V^2$; for the 78-operation equation choose $L = bV$. Set

$$\begin{aligned}
M &= hDH, \qquad R_G = M^2 G, \\
Z &= hD\big[H(n - LB) + K_+\big] - H(DB + he).
\end{aligned}$$

The single equation is

$$\boxed{P_L = Q(Z^2 - R_G V^2) + (R_G - Z^2) = 0.} \tag{13}$$

The circuits use only binary addition, subtraction and multiplication; squares and multiplication by a fixed integer each count as one operation. The sole change from the corresponding parent is $b = cT - R$ in place of $c(T + \alpha) - R$.

The theorem and its proof boundary. For either choice of L, ordinary arithmetic formalizable in ZFC gives an explicit bijection of the positive zero sets of the new and parent polynomials. Composing it with the retained parent calibration gives

$$\mathrm{ZFC} \vdash \Big(\neg\exists \mathbf{x} \in \mathbb{Z}_{>0}^{10}\ P_L(\mathbf{x}) = 0\Big) \longleftrightarrow \mathrm{Con(ZFC)}.$$

The compact branch retains its native compiler and K215 ancestry; it is separate from the direct proof-calculus construction. This statement is an ordinary proof with named inherited dependencies, not a newly emitted end-to-end proof-assistant certificate.

8.2 Why deleting the shift introduces no spurious positive zeros

There is an exact free-polynomial identity

$$P_{\mathrm{new}}(T) = P_{\mathrm{parent}}(T - \alpha).$$

To use it over positive integers we must prove $T > \alpha$ at every new positive zero. It is not valid to assume this inequality as part of the coordinate change.

The unchanged product-square argument works for either sign of V. It gives positive integers g, y, f, I and $\varepsilon \in \{-1, 1\}$ with

$$\begin{gathered} G = g^2, \quad f = 1 + 2a, \quad I = 2Ug, \quad f^2 = F_*, \\ f^2 - \Delta I^2 = 1, \quad Q = (\Delta I)^2, \quad c^2 \mid I, \\ QV^2 - (Q-1)y^2 = 1, \qquad Z/M = \varepsilon g y. \end{gathered}$$

The signed all-index folding lemma recovers $c = \psi_A(p)$ with p odd and positive. Here ψ_A denotes the Pell ordinate for $A^2 - 1$. The first norm gives

$$k = 2\psi_{P_0}(n_0), \qquad P_0 = 2XY^2 + 1.$$

The possibility $n_0 = 1$ would give $c = 2Y + 1$ strictly between 1 and $2A$, so $n_0 \geq 2$. Consequently

$$c > kY > 8XY^3, \qquad c > X, \qquad K_+ > 0.$$

A useful low-index exclusion. Suppose $p = 3$, so $c = 4A^2 - 1$. If $n_0 \geq 3$, then $k > 32X^2Y^4$, whereas $A < 2XY$ gives $c < 16X^2Y^2$; this contradicts $c > kY$. Thus $n_0 = 2$ and $k = 8XY^2 + 4$. The exact identities

$$\begin{aligned} 2c - (X+2)k &= 8Y^2 + 16Y(X+1) - 4X - 2 > 0, \\ (X+3)k - 2c &= 8Y[(X-1)Y - 2(X+1)] + 4X + 6 > 0 \end{aligned}$$

hold for our $X \geq 32, Y \geq 32^3$. Together with $Y < c/k < Y + 1$ they imply $0 < 2Y - X < 3$. This is impossible since $q \mid X, Y$ and $q \geq 32$. Hence $p \geq 5$ and $c > 4\Delta + 3$. The same exclusion works for $X \geq 3, Y \geq 4$ whenever X, Y share any divisor at least three.

A bound that allows both signs. Write $S = \Delta I$. Positive Pell classification gives $S|V| = \chi_S(j)$ and $y = \psi_S(j)$ for $j \geq 1$ odd; even j is impossible modulo S. Also $f^2 > c + 4\Delta + 3$. For $j = 1$, the equation $f^2 b = c \pm 1$ is impossible between consecutive multiples of f^2. For $j = 3$, the equation becomes

$$f^2(b - 4\varepsilon_v \Delta) = c - \varepsilon_v(4\Delta + 3), \qquad \varepsilon_v \in \{-1, 1\}.$$

Its right side is strictly between 0 and f^2 for both signs, using $c > 4\Delta + 3$. Thus $j \geq 5$, and with $v = |V|$,

$$v \geq 16Q^2 - 20Q + 5 > 8Q^2.$$

Since $c < f^2 < Q$ and $f^2 \geq 3$,

$$\begin{aligned} |b| &\geq (v - c)/f^2 > (8Q^2 - Q)/f^2 \\ &= \Delta(1 - f^{-2})(8Q - 1) \geq \tfrac{2}{3}\Delta(8Q - 1) > 4\Delta Q > 4g. \end{aligned}$$

Furthermore V has the sign of b and $v > |b|$. The auxiliary norm implies $y < 3v/2$. Therefore both allowed weights are positive and satisfy

$$gy < L/2.$$

Put $\kappa = K_+/H > 0$, $\rho = e/D$, $t = B/h$, and $W = 1 + hL$. The exact carrier identity is $Z/M = n + \kappa - Wt - \rho$, and $b - e = cT - k + 1 > 0$ gives $\rho < b/D$ even if $b < 0$. If $B \leq -1$, then $Wt \leq -L - 1/h$, whence

$$n = \varepsilon g y - \kappa + Wt + \rho < -L/2 + b/D < 0.$$

For $b < 0$ the final inequality is immediate; for $b > 0$ it follows from $D > 2$ and $L > b$. This contradicts $n > 0$. Thus $B \geq 0$, so $C > 0$ and $\alpha < q$. The fixed mask bounds now give $0 < R < q^4$, while $c > 8XY^3 \geq 8q^{10}$. Consequently $b = cT - R > c/2 > 0$. Finally

$$g^2 = E + \Delta U^2 E^2 > (A - 1)^2 U^2 E^2, \qquad g/c > (A - 1)c \geq q,$$

and therefore

$$T = (b + R)/c > b/c > 4g/c > q > \alpha.$$

The positive inverse coordinate $T_{\text{parent}} = T - \alpha$ is now justified. In the other direction, every positive parent zero maps to a new positive zero by $T_{\text{new}} = T_{\text{parent}} + \alpha$. These maps are inverse bijections of the whole positive zero sets.

Exact arithmetic checks. The entire leading homogeneous components, with $q_0 = \beta J$ and $k_0 = \eta + \zeta$, are

$$\begin{gathered} 2^{14} q_0^{382} w^{46} s^{112} i^{20} k_0^{64} T^4 (q_0 - F - \alpha)^2, \\ -2^{10} q_0^{288} w^{34} s^{84} i^{14} k_0^{46} T^2. \end{gathered}$$

They are nonzero and have degrees 630 and 468. The verifier checks all actual circuit registers through 28 free-polynomial identities per equation, the complete leading forms, the fixed coefficient reconstruction, the three new low-index identities, and ten changed-formula rejection controls per equation. The full proof and exact dependency references are in `compact/PROOF.md`; the incoming ordinary proof and explicit signed folding lemma are included. Finite artifact checks do not prove the unbounded statements by sampling.

8.3 The complete finite coefficient recipe

Every index, set, sum, product and floor below defines a fixed constant, not another unknown. The letter B_* belongs only to this constant recipe; it is distinct from the polynomial abbreviation B above. Put

$$\rho = 2^{67}, \quad B_* = \rho^{8807}, \quad \beta = B_* - 1, \quad G_m(x) = \sum_{j=0}^{m-1} x^j.$$

Define

$$\begin{aligned}\mathcal{E} = {} & \{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1011\} \\ & \cup \{1013, \dots, 1031\} \cup \{1063, 1064, 1095, 2195, 3296\}.\end{aligned}$$

Then

$$\begin{aligned}M_C &= B_* - 1 - \sum_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063}, \\ M_F &= B_* - 1 + 4 + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) \\ &\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\ &\quad + (2^{65} - 1)\rho^{8792} G_{13}(\rho) + (2^{19} - 1)\rho^{8805}, \\ K_0 &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\ &\quad + (\rho^{4393} + \rho^{5495})\Phi(1/\rho),\end{aligned}$$

where

$$\begin{aligned}\Phi(x) &= (1 + 2^{51}) G_{943}(x) - 2^{10} \sum_{i=0}^{942} v(t_i) x^i \\ &\quad + \sum_{(b,e,m) \in \mathcal{C}} 2^b x^e G_m(2x), \\ \mathcal{C} &= \{(10, 1023, 8), (18, 1013, 9), (27, 1003, 8), \\ &\quad (35, 983, 9), (44, 1031, 1), (45, 1022, 1), (46, 1011, 1), \\ &\quad (47, 992, 1), (48, 1095, 1), (49, 2195, 1), (50, 3296, 1)\}.\end{aligned}$$

The rational intermediate in K_0 cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. Let

$$\begin{gathered}h(q, b) = 2(q + 1) + b, \quad \mathcal{R} = \{u_j : d_j = 1,\ 0 \le j < 430\}, \\ \mathcal{L} = \{u_j : d_j = 0,\ 0 \le j < 430\}, \\ r_q = 6 + |\mathcal{R} \cap \{0, \dots, q - 1\}|, \quad l_q = 205 + |\mathcal{L} \cap \{0, \dots, q - 1\}|.\end{gathered}$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are fixed:

index i	tile $t_i = (w, n, e, s)$
$0, 1$	$(2, 435, 3, 1),\ (5, 0, 0, 0)$
$j + 2\ (j = 2q + b < 430)$	$((1 - d_j) l_{u_j}, h(q, b), d_j r_{u_j}, v_j)$
$432 + b\ (b = 0, 1)$	$(0, h(215, b), 0, h(215, b))$
$434 + b\ (b = 0, 1)$	$(0, b, 0, b)$
$424 + 2r_q + b\ (q \in \mathcal{R}, b = 0, 1)$	$(r_q, b, 0, h(q, b))$
$424 + 2l_q + b\ (q \in \mathcal{L}, b = 0, 1)$	$(0, b, l_q, h(q, b))$
$924 + j\ (0 \le j < 4)$	$(0, j + 430\lfloor j/2 \rfloor, 0, 438)$
928	$(2, 434, 2, 0)$
929	$(3, 436, 4, 4)$
930	$(4, 436, 4, 0)$
$497 + t\ (t = 434, 435, 436)$	$(1, 438, 1, t)$
$934+3a+b-6\lfloor a/4 \rfloor$ $a \in \{0,1,4\},\ b = 0,1,2$	$(a, 437, b, 437)$

For $t = (w, n, e, s)$ put

$$v(t) = w + 2^8 n + 2^{17} e + 2^{25} s + 127\, 2^{34}.$$

A closed decoder for the 430 actions. Put $D_b(A, k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430 + q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14(1 - m_q) + m_q D_{16}(E_0, \tau_q),$$
$$u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j) D_{216}(T_0, j - \sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \ldots, 61$, is:

```
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz.
```

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```

The recipe reproduces all four retained integer payloads byte for byte. The largest fixed input coefficient has 592,884 bits. This is a stored-port size, not the coefficient height of the fully expanded polynomial.

9 Reproduction and verification boundaries

The following files specify the result without needing an external repository or a symbolic-mathematics service:

- `equations/P258/equation.json`: every ordered variable and all 186 literal residuals; its adjacent expansion gives every quartic term.
- `equations/P261/`: the complete 261-variable companion.
- `direct/`: the literal guarded source, exact changes, source restorations, translations, arithmetic proof and finite verifier.
- `compact/`: both complete compact circuits, the four fixed integer payloads and their finite reconstruction, the new proof and the retained parent proofs.
- `sources/`: pinned inherited arithmetic and logical interfaces.
- `verify.py`: the combined finite check, including reverse row operations, complete SOS expansion, and parsing the printed residuals.

From the extracted archive root, use Python 3.10 or later:

```
python3 -B verify.py
python3 -B -O verify.py
python3 -B build_endpoints.py --check
python3 -B evaluate.py point.json
```

The default evaluator expects exactly 258 signed integer coordinates. It is an exact evaluator, not a consistency test or a solver. The standalone LaTeX paper is rebuilt with `sh build_report.sh`.

The mathematical claims have two kinds of support. The unbounded arguments prove soundness and completeness, including the delicate order of sign recovery. Finite checks authenticate the entire written polynomial, transformations, degrees, coefficient heights and circuit counts. The latter do not prove an unbounded number-theoretic lemma by sampling. Independent reviews in this packet are separate reviews within this AI-assisted workflow, not outside human peer review.

Formalization is a natural next step. Existing formalized DPRM work shows how arithmetic graphs and proof encodings can be handled by a proof assistant [6]. We do not substitute that general theorem for the explicit coefficients or the calibrated equivalence in this particular construction.

10 What is new, and further research questions

The direct contribution is the completed application of symmetric remainder bounds to the signed C222 graph, including a pre-factorial radix margin, a two-scale merger, and the odd-square recovery order. The compact contribution is the proof that the shift $T + \alpha$ can be removed while preserving the entire positive zero set by an explicit triangular bijection. The row operations give a fully certified coefficient reduction at the new direct endpoints. CRT, Pell equations, ternary nonnegativity representations and elementary row operations are classical ingredients; no historical priority for them is asserted.

Several questions remain particularly concrete:

1. Can all digit levels share one quadratic budget, with fewer added scaling coordinates than the guard coordinates it removes? The adjacent levels work because their ratio is the already present radix q.

2. Can another cofinal arithmetic choice remove one of the two remaining ceiling sign graphs? A proposed restriction must preserve completeness; a convenient square condition cannot simply be assumed.
3. Can the set-theoretic seed and logical vacuity parts of the numerical calculus be compiled separately to reduce the arithmetic front? The repository already contains warnings against dropping necessary vacuity instances. A new front needs its own complete calibration and explicit finite polynomial, not merely a suggestive formula.
4. Can the signed residue-recognition lemma and the revised recovery order be formalized first as reusable Lean/Rocq lemmas, then composed with a verified literal compiler? This would close the principal gap between the current ordinary proof and an end-to-end formal endpoint.
5. Can coefficient-aware row and coordinate transformations lower height further without increasing variables or support? The present 30-operation certificate gives a reproducible starting point, not a lower bound.

References

[1] V. Reshetnikov and contributors, *ProveIt*, immutable comparison snapshot `e0c98c629bf10d30b695884f6b1d2ab361bdf513`. `https://github.com/VladimirReshetnikov/ProveIt/tree/e0c98c629bf10d30b695884f6b1d2ab361bdf513`.

[2] *Cofinal anchors and height one equations for ZFC consistency*, round 17, 25 October 2026, incoming research archive in [1]. Its direct ordinary proof and exact guarded source are retained here.

[3] *Eliminating auxiliary witnesses in Diophantine consistency*, 25 October 2026, `ConZFC_Certificate_Reductions_2026-10-25.zip` in [1]. The compact component is retained byte for byte.

[4] *Pooled low digits and a semantic Pell cap*, `conzfc_graph_digits_2026-10-24.zip` in [1]. The natural-domain pooling argument is a predecessor; the present signed remainder recognition and recovery schedule are different.

[5] J. Bayer, M. David, M. Haßler, Y. Matiyasevich and D. Schleicher, *Diophantine equations over $\mathbb{Z}$: universal bounds and parallel formalization*, arXiv:2506.20909v3, 22 October 2026. `https://arxiv.org/abs/2506.20909`.

[6] J. Bayer, M. David, B. Stock, A. Pal, Y. Matiyasevich and D. Schleicher, *Diophantine Equations and the DPRM Theorem*, Archive of Formal Proofs, 2022. `https://isa-afp.org/entries/DPRM_Theorem.html`.

A Every residual of the main equation

The independent unknowns are exactly $x_0, \dots, x_{257} \in \mathbb{Z}$. The equation is $\sum_{j=0}^{185} r_j^2 = 0$, with the following complete list. All constants below are literal integers; no reference to a proof predicate, variable exponent, recursion, or external numerical port is needed to evaluate the polynomial. The file `equations/P258/variables.tsv` also records the compiler's semantic names, which are not needed for this definition.

$$r_0 = -x_3^2 + x_4$$

$$r_1 = x_0 - x_0x_2 - x_2 + x_9$$

$$r_2 = -1 + x_1 - x_2x_7 - x_3$$

$$r_3 = -1 + x_1 - x_8x_9$$

$$r_4 = 1 + x_4 - x_5x_{20} + x_{18}$$

$$r_5 = -1 + 2x_{10} - x_{10}^2 - 2x_{10}x_{11} + 2x_{11} - x_{11}^2 + x_{21}$$

$$r_6 = -4 - 12x_{10} - 16x_{11} - 4x_{21} + x_{22} - x_{37}$$

$$r_7 = -2 + 2x_{10} - x_{10}^2 - 4x_{10}x_{22} + 4x_{22} - 4x_{22}^2 + x_{23}$$

$$\begin{aligned} r_8 = {} & 67 - 60x_{10} - 3x_{10}^2 - 60x_{10}x_{22} - 24x_{10}x_{23} - 116x_{22} - 124x_{22}^2 \\ & - 64x_{22}x_{23} - 31x_{23} - 4x_{23}^2 + x_{24} \end{aligned}$$

$$r_9 = -2 - x_{11}^2 - 2x_{11}x_{12} - x_{12}^2 + x_{25}$$

$$\begin{aligned} r_{10} = {} & -9 + 7x_{10} - x_{10}^2 - 4x_{10}x_{11} - 4x_{10}x_{25} + 12x_{11} - 4x_{11}^2 - 8x_{11}x_{25} + 12x_{25} \\ & - 4x_{25}^2 + x_{26} \end{aligned}$$

$$r_{11} = -1 + 2x_{10} - x_{10}^2 - 2x_{10}x_{12} + 2x_{12} - x_{12}^2 + x_{27}$$

$$\begin{aligned} r_{12} = {} & 2 - 26x_{10} - 68x_{10}^2 - 24x_{10}x_{11} - 64x_{10}x_{12} + 48x_{10}x_{27} + 48x_{10}x_{30} \\ & + 52x_{11}^2 + 32x_{11}x_{21} + 96x_{11}x_{27} + 64x_{11}x_{30} - 48x_{12} - 16x_{12}^2 + 32x_{12}x_{27} \\ & + 32x_{12}x_{30} + 6x_{21} + 4x_{21}^2 + 24x_{21}x_{27} + 16x_{21}x_{30} - 32x_{27} + 28x_{27}^2 \\ & + 32x_{27}x_{30} + x_{28} - 8x_{30} + 8x_{30}^2 - 2x_{31} \end{aligned}$$

$$r_{13} = -4x_{26}^2 - 8x_{26}x_{28} - 4x_{28}^2 + x_{29}$$

$$\begin{aligned} r_{14} = {} & -6 + 16x_{12} - 16x_{12}^2 - 16x_{12}x_{25} - 16x_{12}x_{27} + 8x_{25} - 4x_{25}^2 - 8x_{25}x_{27} \\ & + 8x_{27} - 4x_{27}^2 + x_{30} \end{aligned}$$

$$\begin{aligned} r_{15} = {} & 27 - 46x_{10} - 10x_{10}^2 - 54x_{10}x_{11} - 38x_{10}x_{12} - 24x_{10}x_{21} - 48x_{10}x_{27} \\ & - 24x_{10}x_{30} - 34x_{11} - 43x_{11}^2 - 64x_{11}x_{12} - 32x_{11}x_{21} - 64x_{11}x_{27} - 32x_{11}x_{30} \\ & - 2x_{12} - 11x_{12}^2 - 16x_{12}x_{21} - 32x_{12}x_{27} - 16x_{12}x_{30} - 19x_{21} - 4x_{21}^2 \\ & - 16x_{21}x_{27} - 8x_{21}x_{30} + 3x_{27} - 16x_{27}^2 - 16x_{27}x_{30} + 4x_{30} - 4x_{30}^2 + x_{31} \end{aligned}$$

$$\begin{aligned} r_{16} = {} & 46 - 5x_{10} - x_{10}^2 - 12x_{10}x_{11} - 16x_{10}x_{12} - 4x_{10}x_{25} - 36x_{11} - 8x_{11}^2 \\ & - 40x_{11}x_{12} - 24x_{11}x_{25} - 48x_{12} - 36x_{12}^2 - 32x_{12}x_{25} - 40x_{25} - 4x_{25}^2 + x_{32} \end{aligned}$$

$$\begin{aligned} r_{17} = {} & 24 - 44x_{10} - 26x_{10}^2 - 12x_{10}x_{11} - 76x_{10}x_{12} - 24x_{10}x_{27} - 3x_{11} + 7x_{11}^2 \\ & - 4x_{11}x_{27} - 52x_{12} - 46x_{12}^2 - 32x_{12}x_{27} - 8x_{25} - 18x_{27} - 4x_{27}^2 + x_{33} \end{aligned}$$

$$r_{18} = 2 - 4x_{32}^2 - 8x_{32}x_{33} - 4x_{33}^2 + x_{34}$$

$$\begin{aligned} r_{19} = {} & -12 + 16x_{10} - 4x_{10}^2 - 8x_{10}x_{12} - 8x_{10}x_{21} - 8x_{10}x_{25} + 16x_{12} - 4x_{12}^2 \\ & - 8x_{12}x_{21} - 8x_{12}x_{25} + 16x_{21} - 4x_{21}^2 - 8x_{21}x_{25} + 16x_{25} - 4x_{25}^2 + x_{35} \end{aligned}$$

$$r_{20} = 22 - 74x_{10} - 39x_{10}^2 - 110x_{10}x_{12} - 64x_{10}x_{25} - 32x_{10}x_{27} - 32x_{10}x_{35} - 76x_{12} - 75x_{12}^2 - 80x_{12}x_{25} - 40x_{12}x_{27} - 40x_{12}x_{35} - 8x_{25} - 16x_{25}^2 - 16x_{25}x_{27} - 16x_{25}x_{35} - 31x_{27} - 4x_{27}^2 - 8x_{27}x_{35} - 4x_{35} - 4x_{35}^2 + x_{36}$$

$$r_{21} = 25 - 42x_{10} - 24x_{10}^2 - 72x_{10}x_{11} - 24x_{10}x_{21} - 48x_{11} - 45x_{11}^2 + 14x_{11}x_{12} - 32x_{11}x_{21} + 7x_{12}^2 - 18x_{21} - 4x_{21}^2 - 7x_{25} + x_{37}$$

$$r_{22} = -2 + 2x_{10} - x_{10}^2 - 4x_{10}x_{37} + 4x_{37} - 4x_{37}^2 + x_{38}$$

$$r_{23} = 67 - 60x_{10} - 3x_{10}^2 - 60x_{10}x_{37} - 24x_{10}x_{38} - 116x_{37} - 124x_{37}^2 - 64x_{37}x_{38} - 31x_{38} - 4x_{38}^2 + x_{39}$$

$$r_{24} = -x_{11}^2 - 4x_{11}x_{39} - 4x_{39}^2 + x_{40}$$

$$r_{25} = 10 - 30x_{10} - 64x_{10}^2 - 128x_{10}x_{11} - 64x_{10}x_{21} - 32x_{11} - 56x_{11}^2 + 16x_{11}x_{12} - 64x_{11}x_{21} + 8x_{12}^2 - 14x_{21} - 16x_{21}^2 - 8x_{25} + x_{41}$$

$$r_{26} = 52 + 8x_{11} - 36x_{11}^2 - 40x_{11}x_{12} - 32x_{11}x_{25} - 32x_{11}x_{41} + 6x_{12} - 8x_{12}^2 - 24x_{12}x_{25} - 24x_{12}x_{41} - 26x_{25} - 4x_{25}^2 - 8x_{25}x_{41} + 4x_{41} - 4x_{41}^2 + x_{42}$$

$$r_{27} = 8x_{10} - 64x_{10}^2 - 96x_{10}x_{12} - 32x_{10}x_{27} - 32x_{10}x_{42} + 6x_{12} - 36x_{12}^2 - 24x_{12}x_{27} - 24x_{12}x_{42} + 2x_{27} - 4x_{27}^2 - 8x_{27}x_{42} + 4x_{42} - 4x_{42}^2 + x_{43}$$

$$r_{28} = -x_{12}^2 - 2x_{12}x_{13} - x_{13}^2 + x_{44}$$

$$r_{29} = 2x_{10} - x_{10}^2 - 4x_{10}x_{12} - 4x_{10}x_{44} + 4x_{12} - 4x_{12}^2 - 8x_{12}x_{44} + 4x_{44} - 4x_{44}^2 + x_{45}$$

$$r_{30} = -26 + 17x_{10} - 3x_{10}^2 - 4x_{10}x_{11} - 4x_{10}x_{25} - 8x_{10}x_{27} + 12x_{11} - 8x_{11}x_{27} + 12x_{25} - 8x_{25}x_{27} - x_{26} + 24x_{27} - 4x_{27}^2 + x_{46}$$

$$r_{31} = -8x_{10} - 12x_{10}^2 - 16x_{10}x_{12} - 32x_{10}x_{27} - 16x_{10}x_{44} - 16x_{10}x_{45} - 16x_{10}x_{46} - 20x_{12} - 32x_{12}x_{27} - 16x_{12}x_{45} - 16x_{12}x_{46} - 16x_{27}^2 - 32x_{27}x_{44} - 16x_{27}x_{45} - 16x_{27}x_{46} - 20x_{44} - 16x_{44}x_{45} - 16x_{44}x_{46} - 6x_{45} - 4x_{45}^2 - 8x_{45}x_{46} - 4x_{46}^2 + x_{47}$$

$$r_{32} = x_{12} - 25x_{12}^2 - 40x_{12}x_{27} - 20x_{12}x_{30} - 16x_{27}^2 - 16x_{27}x_{30} - 4x_{30}^2 + x_{48}$$

$$r_{33} = -1 + 8x_{11} - 4x_{11}^2 - 8x_{11}x_{25} + 8x_{25} - 4x_{25}^2 + x_{49}$$

$$r_{34} = -1 - 16x_{11}^2 - 8x_{11}x_{12} - 32x_{11}x_{25} - 16x_{11}x_{49} - x_{12}^2 - 8x_{12}x_{25} - 4x_{12}x_{49} - 16x_{25}^2 - 16x_{25}x_{49} - 4x_{49}^2 + x_{50}$$

$$r_{35} = 1 + x_{11} - x_{11}^2 - 4x_{11}x_{12} - 4x_{11}x_{50} - 4x_{12}^2 - 8x_{12}x_{50} - 4x_{50}^2 + x_{51}$$

$$r_{36} = -1 - 4x_{51} - 4x_{51}^2 + x_{52}$$

$$r_{37} = -1 - x_{12}^2 - 8x_{12}x_{51} - 4x_{12}x_{52} - 16x_{51}^2 - 16x_{51}x_{52} - 4x_{52}^2 + x_{53}$$

$$r_{38} = 2 + x_{11} - x_{11}^2 - 4x_{11}x_{12} - 4x_{11}x_{53} - 4x_{12}^2 - 8x_{12}x_{53} - 4x_{53}^2 + x_{54}$$

$$r_{39} = -4x_{48}^2 - 8x_{48}x_{54} - 4x_{54}^2 + x_{55}$$

$$r_{40} = -9 + 7x_{10} - x_{10}^2 - 8x_{10}x_{48} - 4x_{10}x_{55} + 24x_{48} - 16x_{48}^2 - 16x_{48}x_{55} + 12x_{55} - 4x_{55}^2 + x_{56}$$

$$r_{41} = -1 - 4x_{56} - 4x_{56}^2 + x_{57}$$

$$r_{42} = -x_{12}^2 - 8x_{12}x_{56} - 4x_{12}x_{57} - 16x_{56}^2 - 16x_{56}x_{57} - 4x_{57}^2 + x_{58}$$

$$r_{43} = -x_{10} - x_{10}^2 - 4x_{10}x_{12} - 4x_{10}x_{58} - 4x_{12} - 4x_{12}^2 - 8x_{12}x_{58} - 4x_{58} - 4x_{58}^2 + x_{59}$$

$$r_{44} = -10 - 12x_{59} - 4x_{59}^2 + x_{60}$$

$$r_{45} = 1 + 4x_{11} - 4x_{11}^2 - 8x_{11}x_{25} - 16x_{11}x_{59} - 8x_{11}x_{60} + 4x_{25} - 4x_{25}^2 - 16x_{25}x_{59} - 8x_{25}x_{60} + 8x_{59} - 16x_{59}^2 - 16x_{59}x_{60} + 4x_{60} - 4x_{60}^2 + x_{61}$$

$$r_{46} = x_{11} - 25x_{11}^2 - 40x_{11}x_{25} - 20x_{11}x_{61} - 16x_{25}^2 - 16x_{25}x_{61} - 4x_{61}^2 + x_{62}$$

$$r_{47} = -10 - 12x_{54} - 4x_{54}^2 + x_{63}$$

$$r_{48} = 1 - 16x_{54}^2 - 16x_{54}x_{62} - 16x_{54}x_{63} - 4x_{62}^2 - 8x_{62}x_{63} - 4x_{63}^2 + x_{64}$$

$$r_{49} = x_{12} - x_{12}^2 - 8x_{12}x_{62} - 4x_{12}x_{64} - 16x_{62}^2 - 16x_{62}x_{64} - 4x_{64}^2 + x_{65}$$

$$r_{50} = -4x_{10} - 4x_{10}^2 - 4x_{10}x_{12} - 8x_{10}x_{21} - x_{12} - x_{12}^2 - 4x_{12}x_{21} - 4x_{21} - 4x_{21}^2 + x_{66}$$

$$r_{51} = 1 + 4x_{10} - 4x_{10}^2 - 8x_{10}x_{21} - 8x_{10}x_{66} + 4x_{21} - 4x_{21}^2 - 8x_{21}x_{66} + 4x_{66} - 4x_{66}^2 + x_{67}$$

$$r_{52} = -2 - 4x_{10} - 16x_{10}^2 - 32x_{10}x_{21} - 16x_{10}x_{65} - 16x_{10}x_{67} - 4x_{21} - 16x_{21}^2 - 16x_{21}x_{65} - 16x_{21}x_{67} - 4x_{65}^2 - 8x_{65}x_{67} - 2x_{67} - 4x_{67}^2 + x_{68}$$

$$r_{53} = -2 - 4x_{68} - 4x_{68}^2 + x_{69}$$

$$r_{54} = -x_3x_{15} + x_{70}$$

$$r_{55} = x_{14} - x_{15} - 2x_{24}x_{70} - x_{71}$$

$$r_{56} = x_{14} - x_{15} - 4x_{26}x_{70} - 2x_{29}x_{70} - x_{72}$$

$$r_{57} = x_{14} - x_{15} - 2x_{31}x_{70} - 6x_{70} - x_{73}$$

$$r_{58} = x_{14} - x_{15} - 4x_{32}x_{70} - 2x_{34}x_{70} - x_{74}$$

$$r_{59} = x_{14} - x_{15} - 2x_{36}x_{70} + 6x_{70} - x_{75}$$

$$r_{60} = x_{14} - x_{15} - 4x_{39}x_{70} - 2x_{40}x_{70} - x_{76}$$

$$r_{61} = x_{14} - x_{15} - 2x_{43}x_{70} - x_{77}$$

$$r_{62} = x_{14} - x_{15} - 2x_{47}x_{70} - 4x_{70} - x_{78}$$

$$r_{63} = -x_{10}x_{11} + x_{10}x_{12} + x_{11} + x_{11}x_{12} - x_{12} - x_{12}^2 + x_{79}$$

$$r_{64} = 1 + 2x_{13} + x_{13}^2 - x_{79}^2 + x_{80}$$

$$r_{65} = -x_{80}^2 + x_{81}$$

$$r_{66} = -4x_{68}x_{70} - 2x_{69}x_{70} + 2x_{70} + x_{82}$$

$$r_{67} = -x_{14}^2 + 2x_{14}x_{15} + 2x_{14}x_{82} - x_{15}^2 - 2x_{15}x_{82} - x_{82}^2 + x_{83}$$

$$r_{68} = -1 - x_3x_{11} + x_{84}$$

$$r_{69} = -x_{15}x_{84} + x_{86}x_{87} + x_{88}$$

$$r_{70} = -1 + x_3 - x_3x_{10} + x_{85}$$

$$r_{71} = -1 + 2x_3 - 2x_3x_{10} - 2x_3x_{21} + x_{86}$$

$$r_{72} = -x_{14}x_{85} + x_{87}$$

$$r_{73} = -1 - 6x_3x_{10} - 8x_3x_{11} - 2x_3x_{21} + x_{89}$$

$$r_{74} = x_{14}x_{84} - x_{15}x_{89} + x_{90}$$

$$r_{75} = -x_{14}x_{71} + x_{15}x_{71} + x_{91}$$

$$r_{76} = -x_{72}x_{91} + x_{92}$$

$$r_{77} = -x_{73}x_{92} + x_{93}$$

$$r_{78} = -x_{74}x_{93} + x_{94}$$

$$r_{79} = -x_{75}x_{94} + x_{95}$$

$$r_{80} = -x_{76}x_{95} + x_{96}$$

$$r_{81} = -x_{77}x_{96} + x_{97}$$

$$r_{82} = -x_{78}x_{97} + x_{98}$$

$$r_{83} = -x_{88}x_{98} + x_{99}$$

$$r_{84} = -x_{90}x_{99} + x_{100}$$

$$r_{85} = -x_{100}^2 + x_{101}$$

$$r_{86} = -x_2 - x_2x_{18} + x_{18} + x_{102}$$

$$r_{87} = -x_{16}x_{102} + x_{103}$$

$$\begin{aligned} r_{88} = {} & -2x_1^2 + 2x_1x_{14} + 2x_1x_{15} + 2x_1x_{103} + 2x_1x_{104} - x_{14}^2 - 2x_{14}x_{103} \\ & - x_{15}^2 - 2x_{15}x_{104} + x_{19}x_{106} - x_{81}x_{101} - x_{83}x_{101} - x_{103}^2 - x_{104}^2 \end{aligned}$$

$$r_{89} = -x_2x_{17} + x_{17} - x_{17}x_{102} + x_{104}$$

$$\begin{aligned} r_{90} = {} & -9x_{10} - 9x_{11} - 4x_{21} - 2x_{24} - 4x_{26} - 2x_{29} - 2x_{31} - 4x_{32} - 2x_{34} - 2x_{36} - 4x_{39} \\ & - 2x_{40} - 2x_{43} - 2x_{47} - 4x_{68} - 2x_{69} + x_{105} \end{aligned}$$

$$r_{91} = -x_{107}x_{110} + x_{110} + x_{112}$$

$$r_{92} = -x_5x_{109} + x_{108} - x_{109}x_{112} - x_{111}$$

$$r_{93} = -x_{19}x_{114} + x_{117}$$

$$r_{94} = x_{113} - x_{115}x_{117} - x_{116}$$

$$r_{95} = -x_{117}x_{119} + x_{118} - x_{120}$$

$$r_{96} = -x_{117}x_{122} + x_{121} - x_{123}$$

$$r_{97} = -x_{117}x_{125} + x_{124} - x_{126}$$

$$r_{98} = -x_{19}x_{128} + x_{131}$$

$$r_{99} = x_{127} - x_{129}x_{131} - x_{130}$$

$$r_{100} = -x_{131}x_{133} + x_{132} - x_{134}$$

$$r_{101} = -x_{19}x_{136} + x_{139}$$

$$r_{102} = x_{135} - x_{137}x_{139} - x_{138}$$

$$r_{103} = -x_{139}x_{141} + x_{140} - x_{142}$$

$$r_{104} = -x_{19}x_{109} + x_{146}$$

$$r_{105} = x_{143} - x_{144}x_{146} - x_{145}$$

$$r_{106} = -x_{117}x_{148} + x_{147} - x_{149}$$

$$r_{107} = 7 + x_{150} - 2x_{150}x_{152} - 14x_{152} + x_{153}$$

$$r_{108} = -x_{150}x_{153} - 8x_{153} + x_{154}$$

$$r_{109} = 2x_{151} + x_{151}^2 + x_{153}^2 - x_{154}^2$$

$$r_{110} = -x_{151}x_{156} - x_{156} + x_{157}$$

$$r_{111} = 2x_{155} + x_{155}^2 + x_{156}^2 - x_{157}^2$$

$$r_{112} = x_4^2 - 2x_4x_{151} - 2x_{151} + x_{158}$$

$$r_{113} = -1 - x_4x_{156} + x_6 - x_{155} - x_{156} + x_{157} + x_{158}x_{159}$$

$$r_{114} = 1 - 2x_5 + x_5^2 - 2x_5x_{151} + x_{160}$$

$$r_{115} = -x_5x_{206} + x_{19} + x_{160}x_{161} - x_{205} + x_{207}$$

$$r_{116} = -1 + 2x_{151}x_{221} - x_{162} + 2x_{221} - x_{221}^2$$

$$r_{117} = -1 + x_{107} + x_{162}x_{163} - x_{205} - x_{206}x_{221} + x_{207}$$

$$r_{118} = -x_{151}x_{165} - x_{165} + x_{166}$$

$$r_{119} = 2x_{164} + x_{164}^2 + x_{165}^2 - x_{166}^2$$

$$r_{120} = 1 - 2x_{107} + x_{107}^2 - 2x_{107}x_{151} + x_{167}$$

$$r_{121} = -1 - x_{107}x_{165} + x_{108} - x_{164} + x_{166} + x_{167}x_{168}$$

$$r_{122} = 4 - 4x_{107} + x_{107}^2 - 2x_{107}x_{151} + 2x_{151} + x_{169}$$

$$r_{123} = -1 - x_{107}x_{156} + x_{109} - x_{155} + x_{156} + x_{157} + x_{169}x_{170}$$

$$r_{124} = 1 + x_{151} - x_{151}x_{172} - x_{172} + x_{173}$$

$$r_{125} = 1 + 2x_{171} + x_{171}^2 - 2x_{172} + x_{172}^2 - x_{173}^2$$

$$r_{126} = -1 + x_{107} - x_{107}x_{172} + x_{113} + x_{167}x_{174} - x_{171} + x_{173}$$

$$r_{127} = -2x_{151} - x_{151}^2 + x_{175}$$

$$r_{128} = 1 - x_{107} - x_{107}x_{155} - x_{107}x_{157} + x_{114} + x_{151}x_{157} + x_{155} - x_{156}x_{175} + 2x_{157} + x_{169}x_{176}$$

$$r_{129} = -x_{151}x_{178} - x_{178} + x_{179}$$

$$r_{130} = 2x_{177} + x_{177}^2 + x_{178}^2 - x_{179}^2$$

$$r_{131} = -1 - x_{107}x_{178} + x_{118} + x_{167}x_{180} - x_{177} + x_{179}$$

$$r_{132} = -x_{151}x_{182} - x_{182} + x_{183}$$

$$r_{133} = 2x_{181} + x_{181}^2 + x_{182}^2 - x_{183}^2$$

$$r_{134} = -1 - x_{107}x_{182} + x_{121} + x_{167}x_{184} - x_{181} + x_{183}$$

$$r_{135} = -x_{151}x_{186} - x_{186} + x_{187}$$

$$r_{136} = 2x_{185} + x_{185}^2 + x_{186}^2 - x_{187}^2$$

$$r_{137} = -1 - x_{107}x_{186} + x_{124} + x_{167}x_{188} - x_{185} + x_{187}$$

$$r_{138} = -x_{151}x_{190} - x_{190} + x_{191}$$

$$r_{139} = 2x_{189} + x_{189}^2 + x_{190}^2 - x_{191}^2$$

$$r_{140} = -1 - x_{107}x_{190} + x_{127} + x_{167}x_{192} - x_{189} + x_{191}$$

$$r_{141} = -x_{151}x_{194} - x_{194} + x_{195}$$

$$r_{142} = 2x_{193} + x_{193}^2 + x_{194}^2 - x_{195}^2$$

$$r_{143} = -1 - x_{107}x_{194} + x_{128} + x_{169}x_{196} - x_{193} + x_{194} + x_{195}$$

$$r_{144} = -x_{151}x_{198} - x_{198} + x_{199}$$

$$r_{145} = 2x_{197} + x_{197}^2 + x_{198}^2 - x_{199}^2$$

$$r_{146} = -1 - x_{107}x_{198} + x_{132} + x_{167}x_{200} - x_{197} + x_{199}$$

$$r_{147} = -x_{151}x_{202} - x_{202} + x_{203}$$

$$r_{148} = 2x_{201} + x_{201}^2 + x_{202}^2 - x_{203}^2$$

$$r_{149} = -1 - x_{107}x_{202} + x_{135} + x_{167}x_{204} - x_{201} + x_{203}$$

$$r_{150} = 1 - x_{151}x_{206} - x_{206} + x_{207}$$

$$r_{151} = -1 + 2x_{205} + x_{205}^2 + x_{206}^2 - 2x_{207} - x_{207}^2$$

$$r_{152} = -x_{107}x_{206} + x_{136} + x_{169}x_{208} - x_{205} + x_{206} + x_{207}$$

$$r_{153} = -x_{151}x_{210} - x_{210} + x_{211}$$

$$r_{154} = 2x_{209} + x_{209}^2 + x_{210}^2 - x_{211}^2$$

$$r_{155} = -1 - x_{107}x_{210} + x_{140} + x_{167}x_{212} - x_{209} + x_{211}$$

$$r_{156} = -x_{151}x_{214} - x_{214} + x_{215}$$

$$r_{157} = 2x_{213} + x_{213}^2 + x_{214}^2 - x_{215}^2$$

$$r_{158} = -1 - x_{107}x_{214} + x_{143} + x_{167}x_{216} - x_{213} + x_{215}$$

$$r_{159} = -x_{151}x_{218} - x_{218} + x_{219}$$

$$r_{160} = 2x_{217} + x_{217}^2 + x_{218}^2 - x_{219}^2$$

$$r_{161} = -1 - x_{107}x_{218} + x_{147} + x_{167}x_{220} - x_{217} + x_{219}$$

$$\begin{aligned} r_{162} = {} & 1 - 2x_0 - 3x_0^2 - 2x_0x_4 - x_1^2 - x_2^2 - 2x_4 - 2x_4^2 - x_5^2 \\ & - x_6^2 + 2x_{10} - x_{10}^2 - x_{11}^2 - x_{12}^2 - x_{13}^2 - x_{14}^2 - x_{15}^2 \\ & - x_{16}^2 - x_{17}^2 - x_{18}^2 - x_{19}^2 - x_{105}^2 + 2x_{107} - 2x_{107}^2 - x_{108}^2 \\ & - x_{109}^2 - x_{113}^2 - x_{114}^2 - x_{118}^2 - x_{121}^2 - x_{124}^2 - x_{127}^2 \\ & - x_{128}^2 - x_{132}^2 - x_{135}^2 - x_{136}^2 - x_{140}^2 - x_{143}^2 - x_{147}^2 + x_{150} \\ & - x_{221}^2 \end{aligned}$$

$$\begin{aligned} r_{163} = {} & -x_5x_{10} - x_5x_{11} - x_5x_{12} - x_5x_{13} - x_5x_{14} - x_5x_{15} - x_5x_{16} \\ & - x_5x_{17} - x_5x_{18} + x_{221} \end{aligned}$$

$$r_{164} = -16 + x_0 - x_{222}^2$$

$$r_{165} = -1 + x_2 + x_6 - x_6^2$$

$$r_{166} = -x_{151}x_{224} - x_{224} + x_{233}$$

$$r_{167} = -1 + x_{223}^2 + x_{224}^2 - x_{233}^2$$

$$r_{168} = -x_{226} - x_{226}x_{227} + x_{234}$$

$$r_{169} = -1 + x_{225}^2 + x_{226}^2 - x_{234}^2$$

$$r_{170} = -x_{218}^2 + x_{235}$$

$$r_{171} = x_{224} - 2x_{229}x_{235} - x_{235}$$

$$r_{172} = -4x_{224} - 8x_{224}x_{228} + x_{227}$$

$$r_{173} = -x_{151} - x_{223}x_{230} + x_{227}$$

$$r_{174} = -1 - x_{217} - x_{223}x_{231} + x_{225}$$

$$r_{175} = -x_{105} - 4x_{218}x_{232} + x_{226}$$

$$\begin{aligned} r_{176} = {} & 2x_0^2 + 2x_0x_4 - 2x_0x_{156} - 2x_0x_{165} + x_1^2 - 2x_1x_{206} + x_2^2 \\ & - 2x_2x_{194} + x_4^2 - 2x_4x_{165} + x_{10}^2 - 2x_{10}x_{172} + x_{11}^2 - 2x_{11}x_{178} \\ & + x_{12}^2 - 2x_{12}x_{182} + x_{13}^2 - 2x_{13}x_{186} + x_{14}^2 - 2x_{14}x_{190} + x_{15}^2 \\ & - 2x_{15}x_{198} + x_{16}^2 - 2x_{16}x_{202} + x_{17}^2 - 2x_{17}x_{210} + x_{18}^2 - 2x_{18}x_{214} \\ & + x_{105}^2 - 2x_{105}x_{218} - x_{151}x_{236} + x_{155}^2 + x_{156}^2 + x_{164}^2 + x_{165}^2 \\ & + x_{171}^2 + x_{172}^2 + x_{177}^2 + x_{178}^2 + x_{181}^2 + x_{182}^2 + x_{185}^2 \\ & + x_{186}^2 + x_{189}^2 + x_{190}^2 + x_{193}^2 + x_{194}^2 + x_{197}^2 + x_{198}^2 \\ & + x_{201}^2 + x_{202}^2 + x_{205}^2 + x_{206}^2 + x_{209}^2 + x_{210}^2 + x_{213}^2 \\ & + x_{214}^2 + x_{217}^2 + x_{218}^2 \end{aligned}$$

$$r_{177} = -1 - x_{3}x_{5} + x_{5} + x_{6} - x_{237}^2 - x_{238}^2 - x_{239} - x_{239}^2$$

$$r_{178} = x_{3}x_{5} - x_{6} - x_{240}^2 - x_{241}^2 - x_{242} - x_{242}^2$$

$$\begin{aligned} r_{179} = {} & -2 - x_{105}^2 - x_{156}^2 - x_{165}^2 + 2x_{172} - x_{172}^2 - x_{178}^2 - x_{182}^2 \\ & - x_{186}^2 - x_{190}^2 - x_{194}^2 - x_{198}^2 - x_{202}^2 - x_{206}^2 - x_{210}^2 \\ & - x_{214}^2 + x_{218}^2 - x_{243}^2 - x_{244}^2 - x_{245} - x_{245}^2 \end{aligned}$$

$$r_{180} = -2 + x_{107} - 4x_{246} - 4x_{246}^2$$

$$r_{181} = -x_{107}x_{111} + x_{111} + x_{247}$$

$$r_{182} = -x_{107}x_{145} + x_{145} + x_{248}$$

$$\begin{aligned} r_{183} = {} & -1 + x_{114}^2 - 4x_{116}^2 - 4x_{120}^2 - 4x_{123}^2 - 4x_{126}^2 - 4x_{149}^2 - 4x_{247}^2 \\ & - 4x_{248}^2 - x_{249}^2 - x_{250}^2 - x_{251} - x_{251}^2 \end{aligned}$$

$$r_{184} = -1 + x_{128}^2 - 4x_{130}^2 - 4x_{134}^2 - x_{252}^2 - x_{253}^2 - x_{254} - x_{254}^2$$

$$r_{185} = -1 + x_{136}^2 - 4x_{138}^2 - 4x_{142}^2 - x_{255}^2 - x_{256}^2 - x_{257} - x_{257}^2$$